

biotechnological inventions intended to exclude products 'exclusively obtained by means of an essentially biological process', the Administrative Council amended R. 27 EPC and R. 28 EPC in 2017 (OJ 2017, A56) to exclude from patentability plants exclusively obtained by means of an essentially biological process.

In T 1063/18, the board, in an enlarged composition consisting of three technically and two legally qualified members, held that R. 28(2) EPC (see OJ 2017, A56) is in conflict with Art. 53(b) EPC as interpreted by the Enlarged Board of Appeal in decisions G 2/12 and G 2/13. The board referred to Art. 164(2) EPC, according to which the provisions of the Convention prevail in case of conflict with the Implementing Regulations, and decided to set the decision under appeal aside and to remit the case to the examining division for further prosecution.

The board did not see any reason to deviate from G 2/12 and G 2/13. The interpretation of the Biotech Directive as put forward in the notice of the European Commission on certain articles of Directive 98/44/EC was not seen as a relevant development because it had not been confirmed in a legally binding way. Considering whether an interpretation of Art. 53(b) EPC, different from that given in decisions G 2/12 and G 2/13, was necessary in view of Art. 31(3)(a) of the Vienna Convention on the Law of Treaties, the board concluded that neither the decision of the Administrative Council to adopt R. 28(2) EPC nor the notice of the European Commission could be regarded as a subsequent agreement between the parties in the sense of the Vienna Convention.

b) G 3/19 – dynamic interpretation of Article 53(b) EPC

On 4 April 2019 the President of the European Patent Office referred to the Enlarged Board of Appeal points of law on the interpretation of Art. 164(2) EPC and the assessment of R. 28(2) EPC under said provision.

In its opinion G 3/19, the Enlarged Board began by analysing the scope and focus of the referral and held that the two issues underlying it were interlinked and could be combined in the following single question:

"Taking into account developments that occurred after a decision by the Enlarged Board of Appeal giving an interpretation of the scope of the exception to patentability of essentially biological processes for the production of plants or animals in Article 53(b) EPC, could this exception have a negative effect on the allowability of product claims or product-by-process claims directed to plants, plant material or animals, if the claimed product is exclusively obtained by means of an essentially biological process or if the claimed process feature define an essentially biological process?"

Interpretation of Art. 53(b) EPC and subsequent agreement or practice

Considering Art. 53(b) EPC on its own, i.e. without reference to R. 28(2) EPC, the Enlarged Board confirmed its earlier decisions in cases G 1/98, G 2/07 and G 1/08, and G 2/12 and G 2/13. Furthermore, the Enlarged Board did not find any subsequent

agreement or practice within the meaning of Art. 31(3)(a) and (b) Vienna Convention regarding its earlier interpretation.

Dynamic interpretation in the light of Rule 28(2) EPC

The Enlarged Board held that the application of the various methods of interpretation provided for in Art. 31 and 32 Vienna Convention, also taking into account the subsequent developments in the contracting states, did not lead to the finding that the term "essentially biological processes for the production of plants" in Art. 53(b) EPC was to be read clearly and unambiguously as extending to products defined or obtained by such processes. It therefore endorsed the respective conclusions that it came to in G 2/12. At the same time, however, the Enlarged Board acknowledged that Art. 53(b) EPC did not prohibit this broader understanding of the process exclusion either. Furthermore, the Enlarged Board recognised that, with the introduction of R. 28(2) EPC, the legal and factual situation underlying decision G 2/12 had substantially changed. This amendment constituted a new aspect that had arisen since the EPC was signed which could give reason to believe that a grammatical, and restrictive, interpretation of the wording of Art. 53(b) EPC conflicted with the legislator's aims, whereas a dynamic interpretation could bring a result that diverged from the wording of the law. The Enlarged Board held that the exclusion of products exclusively obtained by means of an essentially biological process was not incompatible with the wording of Art. 53(b) EPC, which did not rule out this broader construction of the term "essentially biological processes for the production of plants or animals". The Enlarged Board concluded that, in view of the clear legislative intent of the contracting states as represented in the Administrative Council and having regard to Art. 31(4) Vienna Convention, the introduction of R. 28(2) EPC allowed and indeed called for a dynamic interpretation of Art. 53(b) EPC. Accordingly, the Enlarged Board abandoned the interpretation of Art. 53(b) EPC given in decision G 2/12 and, in the light of R. 28(2) EPC, held that the term "essentially biological processes for the production of plants or animals" in Art. 53(b) EPC was to be understood and applied as extending to products exclusively obtained by means of an essentially biological process or if the claimed process feature defined an essentially biological process.

In conclusion, the Enlarged Board of Appeal answered the referral as follows:

"Taking into account developments after decisions G 2/12 and G 2/13 of the Enlarged Board of Appeal, the exception to patentability of essentially biological processes for the production of plants or animals in Article 53(b) EPC has a negative effect on the allowability of product claims and product-by-process claims directed to plants, plant material or animals, if the claimed product is exclusively obtained by means of an essentially biological process or if the claimed process features define an essentially biological process. This negative effect does not apply to European patents granted before 1 July 2017 and European patent applications which were filed before that date and are still pending."

3.4. Microbiological processes and the products thereof

The exception to patentability for plant or animal varieties or essentially biological processes for the production of plants or animals does not apply to microbiological processes or the products thereof (Art. 53(b), second part of sentence, EPC).

3.4.1 The concept of "microbiological processes"

R. 26(6) EPC defines a "microbiological process" as any process involving or performed upon or resulting in microbiological material. The boards have not yet issued a decision interpreting this definition.

According to T. 356/93 (OJ 1995, 545), the concept of "microbiological processes" under Art. 53(b), second part of sentence, EPC refers to processes in which micro-organisms (or their parts) are used to make or to modify products or in which new micro-organisms are developed for specific uses. Consequently, the concept of "products thereof" under Art. 53(b), second part of sentence, EPC encompasses products which are made or modified by micro-organisms as well as new micro-organisms as such. The board defined the term "micro-organism" as including not only bacteria and yeasts, but also fungi, algae, protozoa and human, animal and plant cells, i.e. all generally unicellular organisms with dimensions not visible to the naked eye which can be propagated and manipulated in a laboratory, including plasmids and viruses.

Examining the patentability of the claimed plant grouping, the board in T. 356/93 also addressed the issue whether **multi-step processes** for producing plants which include at least one **microbiological process step** could be considered as a whole to represent "microbiological processes", and whether, consequently, the products of such processes (e.g. plants) might be regarded as being "the products thereof" for the purposes of this provision. The board held that "technical processes including a microbiological step" could not simply be equated with "microbiological processes". Nor could the resulting final products of such a process (e.g. plant varieties) be defined as "products of a microbiological process" within the meaning of the said provision.

3.4.2 Difference between microbiological and genetic-engineering processes

In G. 1/98 (OJ 2000, 111), the Enlarged Board suggested that processes of genetic engineering and microbiological processes are not identical. The term microbiological processes in Art. 53(b) EPC 1973 was used as a synonym of processes using micro-organisms. Micro-organisms are different from the parts of living beings used for the genetic modification of plants. To treat genetically-modified plants as products of microbiological processes within the meaning of Art. 53(b), second part of sentence, EPC 1973, would disregard the purpose of the exclusion of plant varieties in Art. 53(b) EPC 1973, i.e. excluding from patentability subject-matter eligible for protection under the plant breeders' rights system. Therefore, the Enlarged Board took the view that it did not make any difference for the requirements under the UPOV Convention or under the Regulation on Plant Variety Rights, how a variety was obtained. Whether a plant variety was the result of traditional breeding techniques, or whether genetic engineering

was used to obtain a distinct plant grouping, did not matter. This meant that the term "plant variety" was appropriate for defining the borderline between patent protection and plant breeders' rights protection irrespective of the origin of the variety.

4. Medical methods

4.1. Introduction

Art. 53(c) EPC states that European patents shall not be granted in respect of methods for treatment of the human or animal body by surgery or therapy and diagnostic methods practised on the human or animal body; this provision shall not apply to products, in particular substances or compositions, for use in any of these methods. Thus, the exclusion of methods of treatment and diagnostic methods formerly referred to in Art. 52(4) EPC 1973 has been added to the two exceptions to patentability and reappears in the new Art. 53(c) EPC.

In three (almost) identical decisions G 1/83, G 5/83 and G 6/83 (OJ 1985, 60,64, 67), the Enlarged Board stated that the intention of Art. 52(4) EPC 1973 (now Art. 53(c) EPC) was only to prevent non-commercial and non-industrial medical and veterinary activities from being restrained by patent rights (see also T 245/87, OJ 1989, 171). In G 1/04 (OJ 2006, 334) the Enlarged Board of Appeal stated, with reference to diagnostic methods, that their exclusion from patentability seemed actually to be based on socio-ethical and public health considerations. In G 1/07 (OJ 2011, 134), a decision concerning treatment by surgery, the Enlarged Board noted that the exclusion under Art. 52(4) EPC 1973 (Art. 53(c) EPC) had been deliberately maintained by the legislator, thereby confirming the principle that medical and veterinary practitioners' freedom to use the best available treatments to the benefit of their patients, uninhibited by any worry that some treatment might be covered by a patent, should be protected by excluding these activities from patentability.

4.2. Differences and similarities in approaches to establishing whether a method for treatment by surgery or therapy or a diagnostic method involved

In the case law of the boards of appeal a distinction is sometimes made in the approach to methods for treatment by surgery or therapy on the one hand and diagnostic methods on the other. The leading cases are G 1/04 (OJ 2006, 334) concerning diagnostic methods, and G 1/07 (OJ 2011, 134) concerning treatment by surgery.

An important difference between the two approaches is that, whilst a method claim falls under the prohibition on patenting methods for treatment by therapy or surgery if it comprises or encompasses at least **one** feature defining a physical activity or action that constitutes a method step for treatment of a human or animal body by surgery or therapy (G 1/07), a much narrower interpretation of Art. 52(4) EPC 1973 was applied with regard to the exclusion from patentability of diagnostic methods in G 1/04 (see in this chapter I.B.4.6. "Diagnostic methods" – G 1/04).

4.3. No need for practitioner involvement

Referring to G 1/04, the Enlarged Board of Appeal in G 1/07 (a decision concerning treatment by surgery) confirmed that whether or not a method is excluded from patentability under Art. 53(c) EPC cannot depend on the person carrying it out. The Enlarged Board in G 1/07 found that, although the findings in G 1/04 related to diagnostic methods, they dealt quite generally with the exclusion from patentability under Art. 52(4) EPC 1973 and were thus equally valid with respect to the other exclusion conditions contained in the new Art. 53(c) EPC.

In G 1/04 (OJ 2006, 334) the Enlarged Board held that whether or not a method is a diagnostic method may neither depend on the participation of a medical or veterinary practitioner nor on the fact that all method steps can also, or only, be practised by medical or technical support staff, the patient himself or herself, or an automated system (see also G 1/07). This also reflects the fact that technological advances are penetrating human and veterinary medicine and the related professions more and more. Moreover, no distinction is to be made in this context between essential method steps having diagnostic character and non-essential method steps lacking it.

In T 467/18 the board observed that it had been explicitly pointed out in G 1/04 and G 1/07 that the ability to delegate a method at best indicated whether a particular method could be considered to form part of the "core of the medical profession's activities".

In T 1680/08 the claim concerned a non-invasive method and apparatus for optimising the respiration for atelectatic lungs. The appellant submitted that a medical doctor would never be hampered by the claimed method as it was executed by a computer. The board however held that there was no term in Art. 53(c) EPC which would allow concluding that hampering of the practitioner's freedom was a prerequisite for the exclusion to apply in the individual case considered.

4.4. Surgical methods

G 1/07 (OJ 2011, 134) marked an important turning point in the boards' case law on the exclusion of surgical methods from patentability. The Enlarged Board rejected the practice followed by the EPO thus far of broadly construing interventions of a surgical nature, such as in decisions T 182/90 and T 35/99, namely that all methods involving irreversible damage to or destruction of living cells or tissues of the living body were regarded as non-insignificant interventions and thus as surgical treatments, irrespective of the underlying mechanism of the intervention (e.g. mechanical, electrical, thermal, chemical). However, the Enlarged Board did not redefine the term "treatment by surgery", but indicated the direction in which it expected future practice and case law to develop. According to G 1/07, the required new direction is that the exclusion from patentability should not be applied to methods in respect of which the interests of public health, of protection of patients and as a counterpart to that of the freedom of the medical profession to apply the treatment of choice to their patients does not call for their exclusion from patentability.